

2-2-2c

Consider the curve given by $xy^2 - x^3y = 6$.

1. Find the x -coordinate of each point on the curve where the tangent line is vertical.

Part C

1 point is earned for sets denominator of $\frac{dy}{dx}$ equal to 0

1 point is earned for substitutes $y = \frac{1}{2}x^2$ or $x = \pm\sqrt{2y}$ into the equation for the curve

1 point is earned for solves for x -coordinate

Tangent line is vertical when $2xy - x^3 = 0$

$$x(2y - x^2) = 0 \text{ gives } x = 0 \text{ or } y = \frac{1}{2}x^2$$

There is no point on the curve with x -coordinate 0.

$$y = \frac{1}{2}x^2, \frac{1}{4}x^5 - \frac{1}{2}x^5 = 6$$

$$\text{When } -\frac{1}{4}x^5 = 6$$

$$x = \sqrt[5]{-24}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for sets denominator of $\frac{dy}{dx}$ equal to 0

1 point is earned for substitutes $y = \frac{1}{2}x^2$ or $x = \pm\sqrt{2y}$ into the equation for the curve

1 point is earned for solves for x -coordinate

Tangent line is vertical when $2xy - x^3 = 0$

$$x(2y - x^2) = 0 \text{ gives } x = 0 \text{ or } y = \frac{1}{2}x^2$$

There is no point on the curve with x -coordinate 0.

$$y = \frac{1}{2}x^2, \frac{1}{4}x^5 - \frac{1}{2}x^5 = 6$$

$$\text{When } -\frac{1}{4}x^5 = 6$$

$$x = \sqrt[5]{-24}$$

2-2-2c

2-2-2c

2. Find all points in the curve whose x -coordinate is 1, and write an equation for the tangent line at each of these points.
-

Part B

1 point is earned for: $y^2 - y = 6$

1 point is earned for solves for y

2 points are earned for tangent lines

When $x = 1$, $y^2 - y = 6$
 $y^2 - y - 6 = 0$

$$(y - 3)(y + 2) = 0$$

$$y = 3, y = -2$$

$$\text{At } (1, 3), \frac{dy}{dx} = \frac{9-9}{6-1} = 0$$

Tangent line equation is $y = 3$

$$\text{At } (1, -2), \frac{dy}{dx} = \frac{-6-4}{-4-1} = \frac{-10}{-5} = 2$$

Tangent line equation is $y + 2 = 2(x - 1)$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $y^2 - y = 6$

1 point is earned for solves for y

2 points are earned for tangent lines

When $x = 1$, $y^2 - y = 6$
 $y^2 - y - 6 = 0$

$$(y - 3)(y + 2) = 0$$

2-2-2c

$$y = 3, y = -2$$

$$\text{At } (1, 3), \frac{dy}{dx} = \frac{9-9}{6-1} = 0$$

Tangent line equation is $y = 3$

$$\text{At } (1, -2), \frac{dy}{dx} = \frac{-6-4}{-4-1} = \frac{-10}{-5} = 2$$

Tangent line equation is $y + 2 = 2(x - 1)$

2-2-2c

3. Consider the curve given by $xy^2 - x^3y = 6$.

(a) Show that $\frac{dy}{dx} = \frac{3x^2y - y^2}{2xy - x^3}$.

(b) Find all points on the curve whose x -coordinate is 1, and write an equation for the tangent line at each of these points.

(c) Find the x -coordinate of each point on the curve where the tangent line is vertical.

Part A

1 point is earned for the implicit differentiation

1 point is earned for verifies expression for $\frac{dy}{dx}$

$$y^2 + 2xy \frac{dy}{dx} - 3x^2y - x^3 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(2xy - x^3) = 3x^2y - y^2$$

$$\frac{dy}{dx} = \frac{3x^2y - y^2}{2xy - x^3}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the implicit differentiation

1 point is earned for verifies expression for $\frac{dy}{dx}$

$$y^2 + 2xy \frac{dy}{dx} - 3x^2y - x^3 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(2xy - x^3) = 3x^2y - y^2$$

$$\frac{dy}{dx} = \frac{3x^2y - y^2}{2xy - x^3}$$

Part B

1 point is earned for: $y^2 - y = 6$

1 point is earned for solving for y

2 points are earned for tangent lines

2-2-2cWhen $x = 1$, $y^2 - y = 6$

$$y^2 - y - 6 = 0$$

$$(y - 3)(y + 2) = 0$$

$$y = 3, y = -2$$

$$\text{At } (1, 3), \frac{dy}{dx} = \frac{9 - 9}{6 - 1} = 0$$

Tangent line equation is $y = 3$

$$\text{At } (1, -2), \frac{dy}{dx} = \frac{-6 - 4}{-4 - 1} = \frac{-10}{-5} = 2$$

Tangent line equation is $y + 2 = 2(x - 1)$ 

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $y^2 - y = 6$ 1 point is earned for solving for y

2 points are earned for tangent lines

When $x = 1$, $y^2 - y = 6$

$$y^2 - y - 6 = 0$$

$$(y - 3)(y + 2) = 0$$

$$y = 3, y = -2$$

$$\text{At } (1, 3), \frac{dy}{dx} = \frac{9 - 9}{6 - 1} = 0$$

Tangent line equation is $y = 3$

2-2-2c

$$\text{At } (1, -2), \frac{dy}{dx} = \frac{-6 - 4}{-4 - 1} = \frac{-10}{-5} = 2$$

Tangent line equation is $y + 2 = 2(x - 1)$

Part C

1 point is earned for sets denominator of $\frac{dy}{dx}$ equal to 0

1 point is earned for substituting $y = \frac{1}{2}x^2$ or $x = \pm\sqrt{2y}$ into the equation for the curve

1 point is earned for solving for x -coordinate

Tangent line is vertical when $2xy - x^3 = 0$

$$x(2y - x^2) = 0 \text{ gives } x = 0 \text{ or } y = \frac{1}{2}x^2$$

There is no point on the curve with x -coordinate 0.

$$\begin{aligned} \text{When } y = \frac{1}{2}x^2, \frac{1}{4}x^5 - \frac{1}{2}x^5 &= 6 \\ -\frac{1}{4}x^5 &= 6 \\ x &= \sqrt[5]{-24} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for sets denominator of $\frac{dy}{dx}$ equal to 0

2-2-2c

1 point is earned for substituting $y = \frac{1}{2}x^2$ or $x = \pm\sqrt{2y}$ into the equation for the curve

1 point is earned for solving for x -coordinate

Tangent line is vertical when $2xy - x^3 = 0$

$$x(2y - x^2) = 0 \text{ gives } x = 0 \text{ or } y = \frac{1}{2}x^2$$

There is no point on the curve with x -coordinate 0.

$$\begin{aligned}\text{When } y = \frac{1}{2}x^2, \quad \frac{1}{4}x^5 - \frac{1}{2}x^5 &= 6 \\ -\frac{1}{4}x^5 &= 6 \\ x &= \sqrt[5]{-24}\end{aligned}$$

2-2-2c

Consider the curve defined by $2y^3 + 6x^2y - 12x^2 + 6y = 1$.

4. The line through the origin with slope -1 is tangent to the curve at point P . Find the x - and y -coordinates of point P .

Part C

1 point is earned for setting $y = -x$

1 point is earned if substitutes $y = -x$ into equation of curve

1 point is earned if solves for x and y

$y = -x$ is equation of the line.

$$2(-x)^3 + 6x^2(-x) - 12x^2 + 6(-x) = 1$$

$$-8x^3 - 12x^2 - 6x - 1 = 0$$

$$x = -1/2, y = 1/2$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for setting $y = -x$

1 point is earned if substitutes $y = -x$ into equation of curve

1 point is earned if solves for x and y

$y = -x$ is equation of the line.

$$2(-x)^3 + 6x^2(-x) - 12x^2 + 6(-x) = 1$$

$$-8x^3 - 12x^2 - 6x - 1 = 0$$

$$x = -1/2, y = 1/2$$

2-2-2c

5. Write an equation of each horizontal tangent line to the curve.
-

Part B

1 point is earned if sets $\frac{dy}{dx} = 0$

1 point is earned if solves $\frac{dy}{dx} = 0$

1 point is earned if uses solutions for x to find equations of horizontal tangent lines

1 point is earned if verifies which solutions for y yield equations of horizontal tangent lines

$$\frac{dy}{dx} = 0$$

$$4x - 2xy = 2x(2 - y) = 0$$

$$x = 0 \text{ or } y = 2$$

$$\text{When } x = 0, 2y^3 + 6y = 1; y = 0.165$$

There is no point on the curve with y coordinate of 2.

$y = 0.165$ is the equation of the only horizontal tangent line



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned if sets $\frac{dy}{dx} = 0$

1 point is earned if solves $\frac{dy}{dx} = 0$

1 point is earned if uses solutions for x to find equations of horizontal tangent lines

1 point is earned if verifies which solutions for y yield equations of horizontal tangent lines

$$\frac{dy}{dx} = 0$$

$$4x - 2xy = 2x(2 - y) = 0$$

$$x = 0 \text{ or } y = 2$$

2-2-2c

When $x = 0$, $2y^3 + 6y = 1$; $y = 0.165$

There is no point on the curve with y coordinate of 2.

$y = 0.165$ is the equation of the only horizontal tangent line

2-2-2c

6. Consider the curve defined by $2y^3 + 6x^2y - 12x^2 + 6y = 1$.
- (a) Show that $dy/dx = 4x - 2xy / x^2 + y^2 + 1$.
- (b) Write an equation of each horizontal tangent line to the curve.
- (c) The line through the origin with slope -1 is tangent to the curve at point P . Find the x - and y -coordinates of point P .

Part A

1 point is earned for implicit differentiation

1 point is earned if verifies expression for dy/dx

$$6y^2 \frac{dy}{dx} + 6x^2 \frac{dy}{dx} + 12xy - 24x + 6 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(6y^2 + 6x^2 + 6) = 24x - 12xy$$

$$\frac{dy}{dx} = \frac{24x - 12xy}{6x^2 + 6y^2 + 6} = \frac{4x - 2xy}{x^2 + y^2 + 1}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned if verifies expression for dy/dx

$$6y^2 \frac{dy}{dx} + 6x^2 \frac{dy}{dx} + 12xy - 24x + 6 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(6y^2 + 6x^2 + 6) = 24x - 12xy$$

$$\frac{dy}{dx} = \frac{24x - 12xy}{6x^2 + 6y^2 + 6} = \frac{4x - 2xy}{x^2 + y^2 + 1}$$

2-2-2c

Part B

1 point is earned if sets $\frac{dy}{dx} = 0$

1 point is earned if solves $\frac{dy}{dx} = 0$

1 point is earned if uses solutions for x to find equations of horizontal tangent lines

1 point is earned if verifies which solutions for y yield equations of horizontal tangent lines

$$\frac{dy}{dx} = 0$$

$$4x - 2xy = 2x(2 - y) = 0$$

$$x = 0 \text{ or } y = 2$$

$$\text{When } x = 0, 2y^3 + 6y = 1; y = 0.165$$

There is no point on the curve with y coordinate of 2.

$y = 0.165$ is the equation of the only horizontal tangent line



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned if sets $\frac{dy}{dx} = 0$

1 point is earned if solves $\frac{dy}{dx} = 0$

1 point is earned if uses solutions for x to find equations of horizontal tangent lines

1 point is earned if verifies which solutions for y yield equations of horizontal tangent lines

$$\frac{dy}{dx} = 0$$

$$4x - 2xy = 2x(2 - y) = 0$$

$$x = 0 \text{ or } y = 2$$

2-2-2c

When $x = 0$, $2y^3 + 6y = 1$; $y = 0.165$

There is no point on the curve with y coordinate of 2.

$y = 0.165$ is the equation of the only horizontal tangent line

Part C

1 point is earned for setting $y = -x$

1 point is earned if substitutes $y = -x$ into equation of curve

1 point is earned if solves for x and y

$y = -x$ is equation of the line.

$$2(-x)^3 + 6x^2(-x) - 12x^2 + 6(-x) = 1$$

$$-8x^3 - 12x^2 - 6x - 1 = 0$$

$$x = -1/2, y = 1/2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for setting $y = -x$

1 point is earned if substitutes $y = -x$ into equation of curve

1 point is earned if solves for x and y

$y = -x$ is equation of the line.

$$2(-x)^3 + 6x^2(-x) - 12x^2 + 6(-x) = 1$$

$$-8x^3 - 12x^2 - 6x - 1 = 0$$

$$x = -1/2, y = 1/2$$

2-2-2c

7. Let f be the function defined by $f(x) = (1 + \tan x)^{3/2}$ for $-\pi/4 < x < \pi/2$.

Write an equation for the line tangent to the graph of f at the point where $x=0$.

Part A

1 point is earned for $f(0) = 1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$$f(0) = 1$$

$$f'(x) = \frac{3}{2}(1 + \tan(x))^{1/2}(\sec^2(x))$$

$$f'(0) = \frac{3}{2}$$

$$y - 1 = \frac{3}{2}x \quad \text{or} \quad y = \frac{3}{2}x + 1$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for $f(0) = 1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

2-2-2c

Note: max 2/5 for answer only

$$f(0) = 1$$

$$f'(x) = \frac{3}{2}(1 + \tan(x))^{1/2}(\sec^2(x))$$

$$f'(0) = \frac{3}{2}$$

$$y - 1 = \frac{3}{2}x \quad \text{or} \quad y = \frac{3}{2}x + 1$$

2-2-2c

8. Let f be the function defined by $f(x) = (1 + \tan x)^{3/2}$ for $-\pi/4 < x < \pi/2$.
- (a) Write an equation for the line tangent to the graph of f at the point where $x = 0$.
- (b) Using the equation found in part (a), approximate $f(0.02)$.
- (c) Let f^{-1} denote the inverse function of f . Write an expression that gives $f^{-1}(x)$ for all x in the domain of f^{-1} .

Part A

1 point is earned for $f(0) = 1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$$f(0) = 1$$

$$f'(x) = \frac{3}{2}(1 + \tan(x))^{\frac{1}{2}} (\sec^2(x))$$

$$f'(0) = \frac{3}{2}$$

$$y - 1 = \frac{3}{2}x \quad \text{or} \quad y = \frac{3}{2}x + 1$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f(0) = 1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$$f(0) = 1$$

2-2-2c

$$f'(x) = \frac{3}{2}(1 + \tan(x))^{\frac{1}{2}} (\sec^2(x))$$

$$f'(0) = \frac{3}{2}$$

$$y - 1 = \frac{3}{2}x \quad \text{or} \quad y = \frac{3}{2}x + 1$$

Part B

1 point is earned for evaluating their equation from part (a) at $x = 0.02$

$$f(0.02) \approx \frac{3}{2}(0.02) + 1$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for evaluating their equation from part (a) at $x = 0.02$

$$f(0.02) \approx \frac{3}{2}(0.02) + 1$$

Part C

1 point is earned for reversing variables anywhere in process

1 point is earned for solving for $\tan y$

1 point is earned for solving for y

$$x = (1 + \tan y)^{3/2}$$

$$x^{2/3} = 1 + \tan y$$

$$\tan y = x^{2/3} - 1$$

$$y = \arctan(x^{2/3} - 1)$$

OR

$$f^{-1}(x) = \arctan(x^{2/3} - 1)$$

2-2-2c



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for reversing variables anywhere in process

1 point is earned for solving for $\tan y$

1 point is earned for solving for y

$$x = (1 + \tan y)^{3/2}$$

$$x^{2/3} = 1 + \tan y$$

$$\tan y = x^{2/3} - 1$$

$y = \arctan(x^{2/3} - 1)$ OR $f^{-1}(x) = \arctan(x^{2/3} - 1)$

9. Let f be the function defined by $f(x) = \ln(x^2 + 1)$, and let g be the function defined by $g(x) = x^5 + x^3$. The line tangent to the graph of f at $x = 2$ is parallel to the line tangent to the graph of g at $x = a$, where a is a positive constant.



What is the value of a ?

(A) 0.246

(B) 0.430

(C) 0.447

(D) 0.790



2-2-2c

10.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

If g^{-1} is the inverse function of g , write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

Part D

1 point is earned: $g^{-1}(2)$

1 point is earned: $(g^{-1})'(2)$

1 point is earned for the tangent line equation

$$g(1) = 2, \text{ so } g^{-1}(2) = 1$$

$$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned: $g^{-1}(2)$

1 point is earned: $(g^{-1})'(2)$

1 point is earned for the tangent line equation

$$g(1) = 2, \text{ so } g^{-1}(2) = 1$$

$$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$$

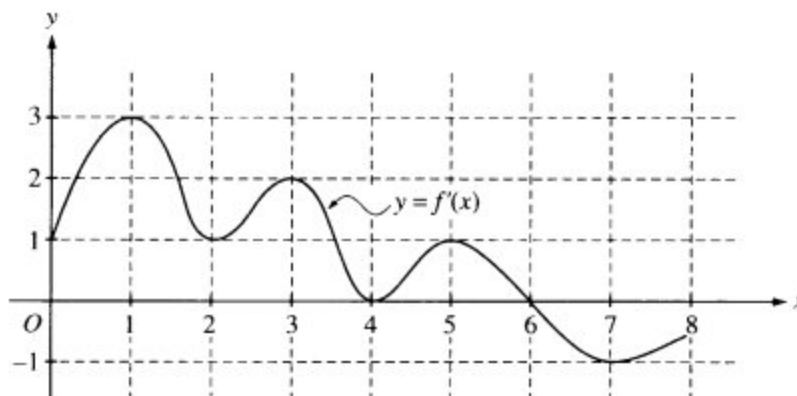
An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

2-2-2c

11. An equation of the line tangent to the graph of $y = x + \cos x$ at the point $(0, 1)$ is
- (A) $y = 2x + 1$
- (B) $y = x + 1$ ✓
- (C) $y = x$
- (D) $y = x - 1$
- (E) $y = 0$
12. An equation of the line tangent to the graph of $y = \frac{2x+3}{3x-2}$ at the point $(1, 5)$ is
- (A) $13x - y = 8$
- (B) $13x + y = 18$ ✓
- (C) $x - 13y = 64$
- (D) $x + 13y = 66$
- (E) $-2x + 3y = 13$
13. Let f be a differentiable function with $f(2) = 3$ and $f'(2) = -5$, and let g be the function defined by $g(x) = xf(x)$. Which of the following is an equation of the line tangent to the graph of g at the point where $x = 2$?
- (A) $y = 3x$
- (B) $y - 3 = -5(x - 2)$
- (C) $y - 6 = -5(x - 2)$
- (D) $y - 6 = -7(x - 2)$ ✓
- (E) $y - 6 = -10(x - 2)$

2-2-2c

14. Refer to the graph and the information below.



The function f is defined on the closed interval $[0, 8]$. The graph of its derivative f' is shown above.

The point $(3, 5)$ is on the graph of $y = f(x)$. An equation of the line tangent to the graph of f at $(3, 5)$ is

(A) $y = 2$

(B) $y = 5$

(C) $y - 5 = 2(x - 3)$

(D) $y + 5 = 2(x - 3)$

(E) $y + 5 = 2(x + 3)$



15. Let f be the function defined by $f(x) = 4x^3 - 5x + 3$. Which of the following is an equation of the line tangent to the graph of f at the point where $x = -1$?

(A) $y = 7x - 3$

(B) $y = 7x + 7$

(C) $y = 7x + 11$

(D) $y = -5x - 1$

(E) $y = -5x - 5$



16.  Which of the following is an equation of the line tangent to the graph of $f(x) = x^4 + 2x^2$ at the point where $f'(x) = 1$?

(A) $y = 8x - 5$

(B) $y = x + 7$

(C) $y = x + 0.763$

(D) $y = x - 0.122$

(E) $y = x - 2.146$



2-2-2c

17. The number of gallons, $P(t)$, of a pollutant in a lake changes at the rate $P'(t) = 1 - 3e^{-0.2\sqrt{t}}$ gallons per day, where t is measured in days. There are 50 gallons of the pollutant in the lake at time $t = 0$. The lake is considered to be safe when it contains 40 gallons or less of pollutant.

An investigator uses the tangent line approximation to $P(t)$ at $t = 0$ as a model for the amount of pollutant in the lake. At what t does this model predict that the lake becomes safe?

Part D

1 point is earned for the slope of tangent line

1 point is earned for the answer

$P'(0) = 1 - 3 = -2$. The lake will become safe when the amount decreases by 10. A linear model predicts this will happen when $t = 5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the slope of tangent line

1 point is earned for the answer

$P'(0) = 1 - 3 = -2$. The lake will become safe when the amount decreases by 10. A linear model predicts this will happen when $t = 5$.

2-2-2c

18. Let $f(x) = 12 - x^2$ for $x \geq 0$ and $f(x) \geq 0$.

The line tangent to the graph of f at the point $(k, f(k))$ intercepts the x -axis at $x = 4$. What is the value of k ?

Part A

1 point is earned for $f'(x) = -2x$

1 point is earned for finding slope of line through $(4, 0)$ & $(k, f(k))$

1 point is earned if equation of tangent line through $(k, f(k))$ or $(4, 0)$ **or** uses other point to get a quadratic

1 point is earned for finding roots of quadratic

1 point is earned for eliminating one answer

$$f(x) = 12 - x^2; f'(x) = -2x$$

slope of tangent line at

$$(k, f(k)) = -2k$$

line through $(4, 0)$ & $(k, f(k))$ has slope

$$\frac{f(k) - 0}{k - 0} = \frac{12 - k^2}{k - 4}$$

$$\text{so } -2k = \frac{12 - k^2}{k - 4} \Rightarrow k^2 - 8k + 12 = 0$$

$$k = 2 \text{ or } k = 6 \text{ but } f(6) = -24$$

so 6 is not in the domain.

$$k = 2$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for $f'(x) = -2x$

2-2-2c

1 point is earned for finding slope of line through $(4,0)$ & $(x,f(x))$

1 point is earned if equation of tangent line through $(k,f(x))$ or $(4,0)$ **or** uses other point to get a quadratic

1 point is earned for finding roots of quadratic

1 point is earned for eliminating one answer

$$f(x) = 12 - x^2; f'(x) = -2x$$

slope of tangent line at

$$(k, f(k)) = -2k$$

line through $(4,0)$ & $(k, f(k))$ has slope

$$\frac{f(k)-0}{k-0} = \frac{12-k^2}{k-4}$$

$$\text{so } -2k = \frac{12-k^2}{k-4} \Rightarrow k^2 - 8k + 12 = 0$$

$$k = 2 \text{ or } k = 6 \text{ but } f(6) = -24$$

so 6 is not in the domain.

$$k = 2$$

2-2-2c

19. Let $f(x) = 12 - x^2$ for $x \geq 0$ and $f(x) \geq 0$.

(a) The line tangent to the graph of f at the point $(k, f(k))$ intercepts the x -axis at $x = 4$. What is the value of k ?

(b) An isosceles triangle whose base is the interval from $(0,0)$ to $(c,0)$ has its vertex on the graph of f . For what value of c does the triangle have maximum area? Justify your answer.

Part A

1 point is earned for $f'(x) = -2x$

1 point is earned for finding slope of line through $(4,0)$ & $(k, f(k))$ **or** equates to slope of tangent line

1 point is earned for write equation of tangent line through $(k, f(k))$ or $(4,0)$ **or** uses other point to get a quadratic

1 point is earned for finding roots of quadratic

1 point is earned for eliminating one answer

$$f(x) = 12 - x^2; f'(x) = -2x$$

slope of tangent line at

$$(k, f(k)) = -2k$$

line through $(4,0)$ & $(k, f(k))$

$$\text{has slope } \frac{f(k) - 0}{k - 4} = \frac{12 - k^2}{k - 4}$$

$$\text{so } -2k = \frac{12 - k^2}{k - 4} \Rightarrow k^2 - 8k + 12 = 0$$

$k = 2$ or $k = 6$ but $f(6) = -24$ so 6 is not in the domain.

$$k = 2$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

2-2-2c

1 point is earned for $f'(x) = -2k$

1 point is earned for finding slope of line through $(4,0)$ & $(k, f(k))$ **or** equates to slope of tangent line

1 point is earned for write equation of tangent line through $(k, f(x))$ or $(4,0)$ **or** uses other point to get a quadratic

1 point is earned for finding roots of quadratic

1 point is earned for eliminating one answer

$$f(x) = 12 - x^2; f'(x) = -2x$$

slope of tangent line at

$$(k, f(k)) = -2k$$

line through $(4,0)$ & $(k, f(k))$

$$\text{has slope } \frac{f(k) - 0}{k - 4} = \frac{12 - k^2}{k - 4}$$

$$\text{so } -2k = \frac{12 - k^2}{k - 4} \Rightarrow k^2 - 8k + 12 = 0$$

$k = 2$ or $k = 6$ but $f(6) = -24$ so 6 is not in the domain.

$$\boxed{k = 2}$$

Part B

$$1 \text{ point is earned for } A = \frac{1}{2}C\left(12 - \frac{C^2}{4}\right)$$

1 point is earned for computing $\frac{dA}{dC}$ & set = 0

1 point is earned for answer

1 point is earned for justification with end points

2-2-2c

$$A = \frac{1}{2}c \cdot f\left(\frac{c}{2}\right) = \frac{1}{2}c \left(12 - \frac{c^2}{4}\right)$$

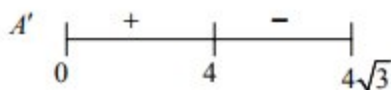
$$= 6c - \frac{c^3}{8} \text{ on } [0, 4\sqrt{3}]$$

$$\frac{dA}{dC} = 6 - \frac{3c^2}{8}; 6 - \frac{3c^2}{8} = 0 \text{ when } c = 4.$$

Candidate test

c	A
0	0
4	16 ← Max
$4\sqrt{3}$	0

First derivative



second derivative

$\frac{d^2A}{dc^2} \Big|_{c=4} = -3 < 0$ so $c = 4$ gives a relative max. $c = 4$ is the only critical value in the domain interval, therefore maximum



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $A = \frac{1}{2}C(12 - \frac{C^2}{4})$

1 point is earned for computing $\frac{dA}{dC}$ & set = 0

1 point is earned for answer

1 point is earned for justification with end points

$$A = \frac{1}{2}c \cdot f\left(\frac{c}{2}\right) = \frac{1}{2}c \left(12 - \frac{c^2}{4}\right)$$

$$= 6c - \frac{c^3}{8} \text{ on } [0, 4\sqrt{3}]$$

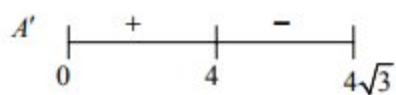
$$\frac{dA}{dC} = 6 - \frac{3c^2}{8}; 6 - \frac{3c^2}{8} = 0 \text{ when } c = 4.$$

2-2-2c

Candidate test

c	A
0	0
4	16 ← Max
$4\sqrt{3}$	0

First derivative



second derivative

$\frac{d^2A}{dc^2}\bigg|_{c=4} = -3 < 0$ so $c = 4$ gives a relative max. $c = 4$ is the only critical value in the domain interval, therefore maximum

2-2-2c

20.

x	2	3	5	8	13
$f(x)$	1	4	-2	3	6

Let f be a function that is twice differentiable for all real numbers. The table above gives values of f for selected points in the closed interval $2 \leq x \leq 13$.

Suppose $f'(5)=3$ and $f''(x)<0$ for all x in the closed interval $5 \leq x \leq 8$. Use the line tangent to the graph of f at $x=5$ to show that $f(7) \leq 4$. Use the secant line for the graph of f on $5 \leq x \leq 8$ to show that $f(7) \geq 4/3$.

Part D

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq \frac{4}{3}$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

2-2-2c

1 point is earned if shows $f(7) \geq \frac{4}{3}$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f'(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f'(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.

21. If the line tangent to the graph of the function f at the point passes through the point $(1, 7)$ passes through the point $(-2, -2)$, then $f'(1)$ is

(A) -5
(B) 1
(C) 3
(D) 7
(E) undefined



22.  Let f be the function given by $f(x) = 3e^{2x}$ and let g be the function given by $g(x) = 6x^3$. At what value of x do the graphs of f and g have parallel tangent lines?

(A) -0.701
(B) -0.567
(C) -0.391
(D) -0.302
(E) -0.258



2-2-2c

23. The function g is defined by $g(x) = x^2 + bx$, where b is a constant. If the line tangent to the graph of g at $x = -1$ is parallel to the line that contains the points $(0, -2)$ and $(3, 4)$, what is the value of b ?
- (A) -1
(B) 2
(C) $\frac{5}{2}$
(D) 4 ✓

Answer D

This option is correct. Parallel lines have the same slope. The slope of the tangent line at $x = -1$ is therefore equal to the slope of the line through $(0, -2)$ and $(3, 4)$, which is $\frac{4 - (-2)}{3 - 0} = 2$. Therefore find b so that

$$g'(-1) = 2$$

$$g'(x) = 2x + b$$

$$g'(-1) = 2(-1) + b = 2 \Rightarrow b = 4$$

24.  Line L is tangent to the graph of $y = \cos x$ at the point $(k, \cos k)$ where $0 < k < \pi$. For what value of k does line L pass through the origin?
- (A) 0.860
(B) 1.571
(C) 2.356
(D) 2.798 ✓
(E) There is no such value of k .

2-2-2c

25. Let f be the function defined by $f(x) = ax^2 + bx$, where a and b are nonzero constants. If the line tangent to the graph of f at the point $(1, -7)$ has slope -5 , what is the value of b ?
- (A) -19
(B) -14
(C) -9 ✓
(D) -2

Answer C

Correct. The values of a and b can be found by setting up two equations using the coordinates of the point of tangency and the slope of the tangent line at $x = 1$.

$$f(1) = a + b = -7$$

$$f'(1) = 2a + b = -5$$

The solution to this system of equations is $a = 2$ and $b = -9$.

26. What is the slope of the line tangent to the graph of $y = \ln(2x)$ at the point where $x = 4$?
- (A) $\frac{1}{8}$
(B) $\frac{1}{4}$ ✓
(C) $\frac{1}{2}$
(D) $\frac{3}{4}$
(E) 4
27. What is the slope of the line tangent to the graph of $y = \ln(2x)$ at the point where $x = 4$?
- (A) $\frac{1}{8}$
(B) $\frac{1}{4}$ ✓
(C) $\frac{1}{2}$
(D) $\frac{3}{4}$
(E) 4